

Chapter 6

Rate of Return Analysis: Single Alternative

The most commonly quoted measure of economic worth for a project or alternative is its Rate of Return (ROR). Compared to the PW or AW value, the ROR is a fundamentally different type of measure of worth, as is discussed in this chapter. The ROR is known by other names such as the *internal rate of return* (IROR), which is the technically correct term, *return on investment* (ROI), *investor's method*, *discounted cash-flow method*, and the *profitability index*.

6.1 What is Rate of Return

Assume the purchase of a 3D printer for \$1,000. By producing and selling manufactured goods, the printer generates an annual revenue of \$300 for the next four years. The ROR

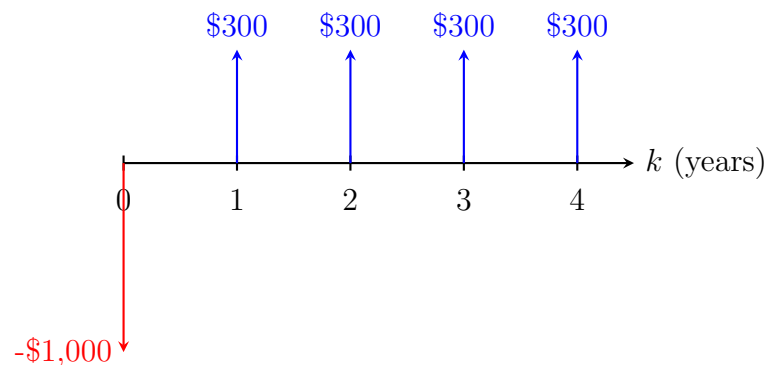


Figure 6.1: Cash flow diagram for the 3D printer investment.

is defined as the interest rate i^* that brings the Present Worth (PW) of the cash flow to

zero. This is formally expressed as:

$$PW(i^*) = 0$$

For this problem, i^* is found by evaluating:

$$PW(i^*) = -P + A(P/A, i^*, 4) = 0$$

This yields $i^* = 7.71\%$. This is further illustrated in Figure 6.2.

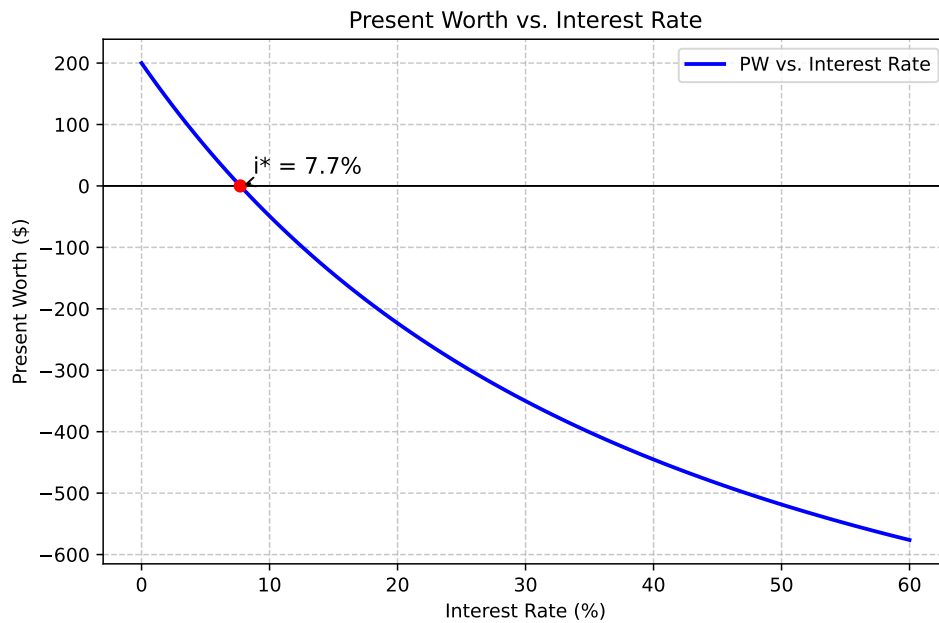


Figure 6.2: Present Worth versus Interest Rate for the 3D printer investment.

To understand the formal definition of ROR, we can examine the unrecovered balance of the 3D printer investment over time.

Definition – Unrecovered Balance

The unrecovered balance is the remaining amount of the initial investment that has not yet been recovered by the cash flows, plus the interest earned on that balance.

Using the calculated $i^* = 7.71\%$, we can track this balance year by year:

As shown in Table 6.1, the final cash flow exactly recovers the remaining balance and its accumulated interest, bringing the ending balance to **zero**. This leads us to the formal definition of the Rate of Return.

Year	Beginning Unrecovered Balance	Interest on Unrecovered Balance (7.71%)	Cash Flow	Recovered Amount	Ending Unrecovered Balance
0	\$0.00	\$0.00	-\$1000.00	—	-\$1000.00
1	-\$1000.00	-\$77.14	\$300.00	\$222.86	-\$777.14
2	-\$777.14	-\$59.95	\$300.00	\$240.05	-\$537.09
3	-\$537.09	-\$41.43	\$300.00	\$258.57	-\$278.52
4	-\$278.52	-\$21.48	\$300.00	\$278.52	\$0.00

Table 6.1: Unrecovered balance of the 3D printer investment over 4 years.

Definition – Rate of Return

Rate of return (ROR) is the rate of interest paid on the unrecovered balance of an investment, so that the final receipt brings the balance to exactly zero with interest considerations.

Important

The Rate of Return (ROR) is **not** the interest rate earned on the original investment amount.

It is strictly the rate earned on the unrecovered balance over time. Calculating interest on the initial principal for the entire duration is a different concept, typically associated with consumer instalment financing.

Note – Similarity to Amortization Schedules

This calculation is mathematically identical to a loan amortization schedule seen in Chapter 2. Instead of paying off a loan, the cash flows from the project are “paying off” the initial investment. The rate i^* is the exact interest rate that makes this “loan” perfectly amortize over the project’s life.

6.2 ROR Calculation

As we have previously expressed, the ROR is the interest rate that makes the PW or AW of a cash flow exactly equal to zero. Hence, i^* can be found using either:

$$PW(i) = \sum_{t=0}^n F_t(1+i)^{-t} = 0 \quad (6.1)$$

$$PW_D = PW_R \quad (6.2)$$

where F_t is the net cash flow at time t , and the subscripts “D” and “R” denote disbursements (costs) and receipts, respectively. Similarly, i^* can be found using the AW :

$$\begin{aligned} AW &= 0 \\ AW_D &= AW_R \end{aligned} \tag{6.3}$$

Notice that Equation 6.1 is an n -th degree polynomial. Solving for i^* involves finding the root(s) of this polynomial. This is solved by trial and error, or using a computer package such as Excel.

Value of i^* The numerical value of the ROR can range from -100% to infinity (i.e., $-100\% \leq i^* \leq \infty$). A value of -100% indicates that the entire invested amount is lost.

Decision-Making The purpose of engineering economy is to determine whether an alternative is economically viable. From the definition of ROR (and from Equations 6.2 and 6.3), the value of ROR ensures that the cash inflows and outflows are equivalent. Consequently, the decision that the project is economically viable or not follows:

Important – Project Evaluation

- $i^* \geq MARR \Rightarrow$ accept alternative as economically viable.
- $i^* < MARR \Rightarrow$ reject the project as economically not viable.

Advantages of ROR The ROR method provides several practical benefits for project evaluation:

- It does not require estimating a MARR beforehand.
- The percentage format is highly intuitive and favored by investors.

Disadvantages of ROR However, this method also presents some distinct analytical challenges:

- It can be computationally difficult to solve for complex cash flows.
- Certain cash flow patterns can yield multiple i^* values, causing ambiguity (see Section 6.3).

6.2.1 Manual Solution Procedure

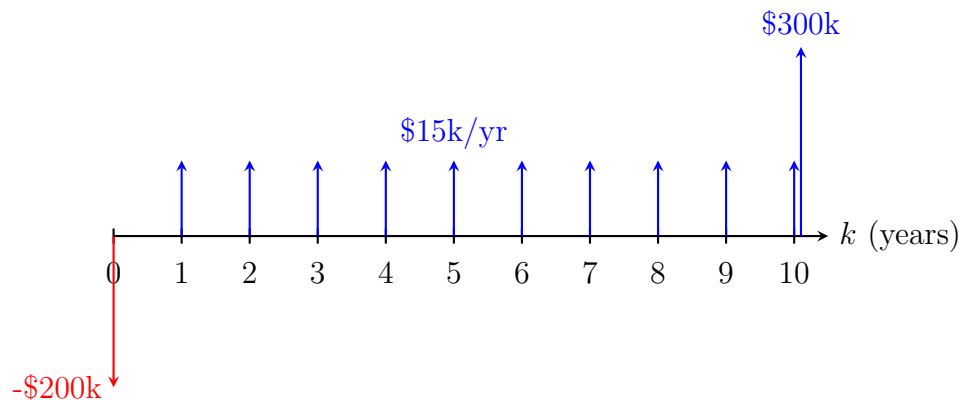
As highlighted, ROR is found, when solved by hand, by trial and error. The procedure is as follows:

1. Convert all disbursements into **either** a single **or** uniform amount **without** considering time value of money.
2. Convert all receipts into **either** a single **or** uniform amounts.
3. Depending on the conversion, solve the resulting $PW(i) = 0$. That is, solve:

$$PW_D = PW_R$$

4. Once an initial guess is estimated, refine and interpolate.

Example 6.1. Engineers at Colorful Paints recommend an investment of \$200,000 now in sustainable manufacturing methods. This investment will reduce waste and generate estimated savings of \$15,000 per year for the next 10 years. Additionally, there will be a one-time savings of \$300,000 at the end of 10 years in facility upgrade costs. Determine the rate of return (i^*) for this investment using manual estimation.



Solution:

1. The single disbursement is the initial investment at $t = 0$:

$$PW_D = \$200,000$$

2. The receipts consist of an annual uniform series and a single future amount at

$t = 10$. Their present worth is:

$$PW_R = 15,000(P/A, i, 10) + 300,000(P/F, i, 10)$$

3. Use the estimation procedure to determine i for the first trial. All income will be regarded as a single F in year 10 so that the P/F factor can be used.

Important – Importance of What to Transform

The P/F factor is selected because most of the cash flow (\$300,000) already fits this factor, and errors created by neglecting the time value of the remaining money will be minimized. Temporarily ignoring the time value of money provides a rough boundary for our initial guess before we refine it.

Only for the first estimate of i , define $P = \$200,000$, $n = 10$, and $F = 10(15,000) + 300,000 = \$450,000$. Now we can state that:

$$\begin{aligned} PW_D &= PW_R \\ P &= F(P/F, i, 10) \\ 200,000 &= 450,000(P/F, i, 10) \\ (P/F, i, 10) &= 0.444 \end{aligned}$$

The roughly estimated i is between 8% and 9%. Use 9% as the first trial because this approximate rate for the P/F factor will be lower than the true value when the time value of money is considered.

4. Trial 1 at $i = 9\%$:

$$\begin{aligned} PW &= -200,000 + 15,000(P/A, 9\%, 10) + 300,000(P/F, 9\%, 10) \\ PW &= \$22,986 > 0 \end{aligned}$$

The result is positive, indicating that the return is more than 9%. Try $i = 11\%$.

5. Trial 2 at $i = 11\%$:

$$\begin{aligned} PW &= -200,000 + 15,000(P/A, 11\%, 10) + 300,000(P/F, 11\%, 10) \\ PW &= -\$6,002 < 0 \end{aligned}$$

6. Since the interest rate of 11% is too high, linearly interpolate between 9% and 11%:

$$\begin{aligned} i^* &= 9.00\% + \frac{22,986 - 0}{22,986 - (-6002)}(2.0\%) \\ &= 9.00\% + 1.59\% = 10.59\% \end{aligned}$$

Note – What if we used the P/A factor?

If we instead transform the \$300,000 future amount into an annual amount by ignoring the time value of money, we get $A = \$15,000 + (\$300,000/10) = \$45,000$. Then, $(P/A, i, 10) = 200,000/45,000 = 4.444$, which corresponds to an estimated return of $i = 18.31\%$. This is very far from the true 10.59% rate of return.

6.2.2 Solution by Excel

In Excel, there are two primary functions to determine the Rate of Return depending on the cash flow pattern. If the cash flow consists of a uniform series (A) over n periods, along with a separate initial investment (P) and final future value (F), use the **RATE** function. The syntax for this single-cell function is **=RATE(n, A, P, F)**. However, if the net cash flows vary from period to period, you must enter them sequentially into a contiguous range of cells, ensuring you include any \$0 amounts.

For these varying cash flows, use the **IRR** function with the syntax **=IRR(first_cell:last_cell, [guess])**. The optional “guess” argument provides a starting point for Excel’s search algorithm to find i^* . This argument does not affect the solution for cash flows having a single root. However, when a cash flow has multiple ROR values, the initial “guess” leads to a different result for the **IRR** function.

6.3 Multiple Rate of Return Values

Because finding the ROR is equivalent to finding the roots of an n -th degree polynomial, it is mathematically possible for the function to have multiple roots.

There are two tests relying on this idea: Descartes’ rule of signs and Norstrom’s criterion.

Definition – Descartes’ rule

The **maximum** number of roots where $i > -100\%$ is equal to the number of sign changes in the cash flows.

For instance, the maximum number of roots for the cash flow in Figure 6.3 is 2. Drawing the $PW(i)$ reveals that this cashflow has indeed 2 roots, one at around 5% and the second at around 24%, as shown in Figure 6.5. Alternatively, the cash flow in Figure 6.4 has 3 sign changes. This cash flow has 3 roots, but only one of them is positive ($i > 0\%$).

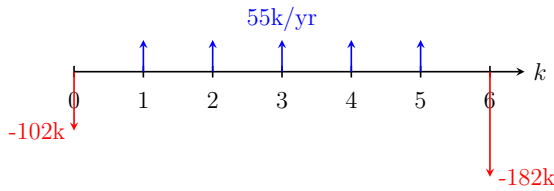


Figure 6.3: Two sign changes.

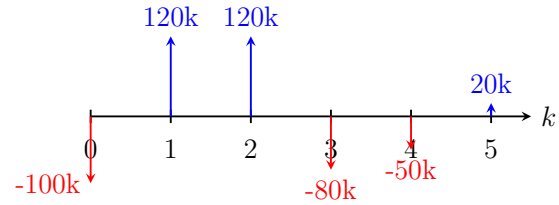
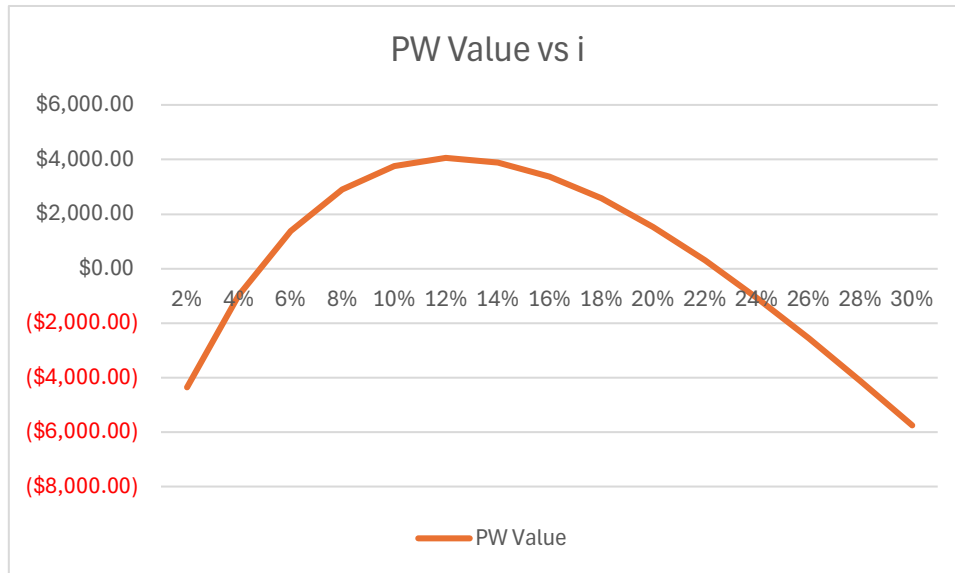


Figure 6.4: Three sign changes.

Figure 6.5: $PW(i)$ for the cashflow in Figure 6.3 highlighting the multiple roots

Definition – Norstrom's criterion

Only one sign change in a series of cumulative cash flows which starts negatively indicates that there is a unique **positive rate of return** ($i > 0\%$).

Let $S_t = \sum_{r=0}^t F_r$ be the cumulative cash flows at time t . The cash flow series has a unique positive ROR if and only if:

- $S_0 < 0$
- The series $(S_0, S_1, \dots, S_n)$ changes sign only once (from minus to plus)

Example 6.2 (Applying Norstrom's Criterion). Let's evaluate the two previous cash flows using Norstrom's criterion by calculating their cumulative cash flows (S_t):

Year (t)	S_t (Figure 6.3)	S_t (Figure 6.4)
0	-\$102k	-\$100k
1	-\$47k	+\$20k
2	+\$8k	+\$140k
3	+\$63k	+\$60k
4	+\$118k	+\$10k
5	+\$173k	+\$30k
6	-\$9k	—

For the first cash flow, although $S_0 < 0$, the series changes sign twice (at year 2 and year 6). Because it changes sign more than once, Norstrom's criterion is not satisfied, confirming why this cash flow has multiple roots.

For the second cash flow, $S_0 < 0$ and the series changes sign exactly once (at year 1). This perfectly satisfies Norstrom's criterion, mathematically guaranteeing a unique positive rate of return.

Note – Norstrom's Criterion vs. Descartes' Rule

This example highlights the analytical power of Norstrom's criterion over Descartes' rule. Even when Descartes' rule is inconclusive due to multiple sign changes, Norstrom's criterion can still definitively prove the existence of a unique positive rate of return.

Important – Guaranteeing Uniqueness

A cash flow does **not** need to satisfy both Descartes' rule and Norstrom's criterion to guarantee a unique positive rate of return. Satisfying **either** test (having exactly one sign change in the raw cash flows **or** exactly one sign change in the cumulative cash flows) is entirely sufficient on its own.

Important – Guidelines for Dealing with Multiple ROR Values

The ROR Values are	Action
Both $i^* < 0$	Discard both
Both $i^* > 0$	Discard both
One $i^* < 0$ & One $i^* > 0$	Use positive i^* as ROR

Table 6.2: Guidelines for Dealing with Multiple ROR values

6.3.1 Conventional and Unconventional Cash Flow

A cash flow series is classified based on the number of sign changes it contains over its lifetime. A series is considered **conventional** if its cash flows change sign exactly once (typically from negative to positive). Because they have only one sign change and begin with a negative investment, **conventional cash flows are guaranteed to have a unique Rate of Return.**

If the cash flow changes sign more than once, the series is classified as **unconventional**. Unconventional cash flows can **potentially** yield multiple ROR values. Currently, there is no universally accepted standard for analysing unconventional cash flows with multiple roots. However, several advanced multi-ROR analysis methods rely on assuming a specific reinvestment rate for any positive cash flows generated during the project's life.

Below is a summary comparing the two types of cash flows:

Feature	Conventional	Unconventional
Sign Changes	Exactly one (usually – to +)	More than one
Number of RORs	Unique positive rate of return	Potentially multiple roots
Analysis Approach	Standard ROR method	Requires reinvestment rate assumptions

Table 6.3: Comparison of conventional and unconventional cash flows.

6.4 Summary

This chapter introduced the Rate of Return (ROR) for evaluating a single investment alternative. The ROR is the interest rate that equates the present worth of inflows to outflows. Unconventional cash flows can yield multiple ROR values and require careful analysis. Evaluating a single project is a crucial first step. However, engineering decisions typically involve choosing among several competing options. Subsequent chapters will build upon these concepts to compare multiple mutually exclusive alternatives.